Pratyusha Kumar

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EDUCATION

B.S. Computer Science Magna Cum Laude
Arizona State University, Tempe, AZ
M.S. Robotics and Autonomous Systems (AI) Dec 2026
Arizona State University, Tempe, AZ

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, MATLAB, MySQL, HTML, CSS, JavaScript, sqlite3, XML
Framework/Libraries: React.JS, Django, Flask, JavaFX, JUnit, Scapy, Pandas, .NET, PyTorch, Tensorflow
Tools: Microsoft Office, Power BI, Microsoft Azure, Git, Jira, Alteryx, Azure DevOps, Taiga, AWS (EC2, S3, SQS, Lambda)

EXPERIENCE

AAIR Research Volunteer June 2026 - Present
- Ported the demonstration collection pipeline from a published AAIR lab paper from OpenRAVE to PyBullet using Python
- Integrating a Vision-Language-Action (VLA) model into the lab's lifted action motion planner pipeline
- Collaborating with lab researchers to adapt and extend existing robot planning code for simulation-based experiments

ARED - AI Agent Feb 2025 - Dec 2025
- Designed a custom preprocessing workflow that reduced compute usage by 30%, achieving 5.3s avg response times
- Built a multi-model AI architecture leveraging lightweight classification and NER models to gate LLM invocation, improving overall efficiency and end-to-end latency using Python
- Deployed the fully containerized, local AI infrastructure using K3s, hosting model weights and inference servers on-device to support business and customer-facing interactions (function calling, translation, and STT) on an edge system

Tech Mahindra, New Delhi, India (Trainee) Apr 2023 - Jun 2023
- Practiced using external databases to collect form data and perform basic visual analysis
- Created form prototypes for client applications and upskilled front-end skills by using React

PROJECTS

Finding Potential Joins in Datasets using LLMs Feb 2025 - Dec 2025
- Collaborated with 3 engineers to build a column-level embedding pipeline for structured datasets using Python
- Stored embeddings in an ANN index to enable fast semantic column retrieval from natural-language queries
- Implemented LLM-based re-ranking to refine top similar columns identified by embedding similarity
- Evaluated multiple embedding models, ANN index formats, and LLMs to identify the most accurate and efficient pipeline configuration

LLM-Controlled Robotic Arm Apr 2026 - May 2026
- Integrated a Gemini LLM with a Dobot Magician Lite robot to execute pick-and-place operations from natural language instructions using Python and pydobot
- Implemented camera-based ground truth detection for real-time spatial awareness and accurate task execution

SASE Website (Webmaster) Dec 2023 - Aug 2024
- Trained and managed a team of 4 developers to update the existing site and its features
- Created a detailed tracking system that tracks student details, emails, attendance with the use of Django and Python
- Increased efficiency by 15% along with 160 students and an average monthly traffic of 3500+ users using the website to sign in to events and for event information

Bridge2Africa E-Learning Lab Jan 2023 - Mar 2024
- Led a team of engineers to document and modernize the website for our client (Bridge2Africa) by updating the UI/UX
- Learned how to create an efficient work environment for a team, assign tasks and responsibilities that lead to productive outcomes, and clearly communicate with a client

EXTRACURRICULAR ACTIVITIES

Society of Asian Scientists and Engineers, ASU Aug 2024 - Present
Director of Internal Affairs, Vice President
- Act as a leader for a team of 9 students to host social, and professional events to student experience
- Perform duties such as presenting slides, reviewing resumes, conducting mock interviews, and conflict management
- Engage with industry representatives and host industry events for ASU students